

CS8395-04: AI-Assisted Design for Cyber Physical Systems (Fall 2026)

Instructor: Sandeep Neema

Email: sandeep.neema@vanderbilt.edu

Office: 301C, 1025 16th Ave. South

Office Hours (TBA): <https://calendly.com/sandeep-neema/15min>

Credits: 3 credit hours, 150 minutes/week

Classroom: FGH 129

Class Time: 4:15 – 5:30, Tuesdays and Thursdays

Course Site: Brightspace (lecture materials, readings, and assignment submissions)

Course Description: Accelerating advances in AI, including Large Language Models (LLMs), Deep Reinforcement Learning (DRL), and Physics-Informed Neural Networks (PINNs) are collectively revolutionizing design automation. The state of practice in design automation for Cyber Physical Systems is Model-based Systems Engineering (MBSE), where designs are manually constructed and represented in myriads of domain-specific modeling languages & tools (Simulink, Modelica, UML, SysML, CAD); designs are evaluated in foundational analysis methods (discrete event simulations, differential algebraic equation solvers, finite element methods); and designs are optimized through multi-disciplinary analysis and optimization methods. This state of practice is rapidly being transformed with new AI-based methods. Generative design techniques and LLMs are being applied for design construction, DRL is being used for design exploration and optimization, while PINNs are being used for accelerating design analysis and evaluation. In this seminar course, we will conduct a review of model-based design automation, explore AI-based advances in design automation, and identify opportunities for impactful research and development in this rapidly advancing field.

Learning Objectives: Upon successful completion of this course, students will be able to:

- explain key concepts and tools in model-based design automation for CPS (MBSE, SysML, architecture, and design space modeling);
- understand pathways to analyze CPS behavior using hybrid (ODE) and higher-fidelity (PDE/FEA/CFD) methods, and reason about multi-disciplinary tradeoffs and optimization (MDAO)
- understand capabilities and limitations of current model-based design processes
- apply generative and predictive AI methods – surrogate models, PINNs, AI in DSE, LLMs, and agentic frameworks – to scale and accelerate design automation
- identify and articulate opportunities for impactful research and development at the intersection of AI and CPS design

Course Format: This is a special-topics course with an exploratory flavor, in an area evolving rapidly through active research and development in academia, established industry, and startups. Students should think of the course as a curated and guided exploration. It introduces key topics and concepts in AI-assisted design of CPS through lectures and invites hands-on exploration of tools and methodologies. Class sessions combine short lectures with student paper/tool presentations and discussion; active participation is expected.

Books: There will not be a textbook for the course. All content will be delivered via lectures, in-class discussions, or outside readings. Readings will primarily consist of recent research papers, review articles, and relevant online resources.

Lectures: Lectures will be delivered in-person and students are responsible for being present in the class, and participating in the discussions. All lecture material will be posted.

Pre-requisites & Intended Audience: As a special-topics seminar, the course is open to both graduate and advanced undergraduate students. The course draws on several areas but introduces domain-specific concepts as needed; students are expected to bring general technical readiness, systemic thinking, and intellectual curiosity rather than deep prior expertise in every topic.

Honor Code: The Vanderbilt Honor Code will govern all course assessments. Students are expected to familiarize themselves with the Honor Code: <https://studenthandbook.vanderbilt.edu/community-standards>. ANY VIOLATIONS will result in the case being reported to the Honor Council. You are welcome to refer to the online sources for your assignments; however, you must not copy the code and must provide citation of the source of inspiration if you refer to online sites. If you have questions regarding what is or is not acceptable, please ask.

Generative AI: You are encouraged to use generative AI tools. Given the breadth of the topics covered and absence of prerequisites, we expect that these tools would be a good aid to learn and explore unfamiliar topics. However, the value you derive from this course is predicated upon you understanding the concept, not a large language model (LLM). In keeping with the Honor Code, please acknowledge any substantial use of generative AI tools in your submitted work, and ensure that what you submit reflects your own understanding.

Grading:

- **Presentations:** 30% (papers, tools tied to the weekly topics)
- **Mini Project:** 30% (model-based design automation focus, report)
- **Final Project:** 40% (AI-assisted design automation focus, report, recorded presentation)

Grade Distribution:

<u>Score</u>	<u>Letter</u>
>= 93.00	A
90.00 - 92.99	A-
87.00 - 89.99	B+
83.00 - 86.99	B
80.00 - 82.99	B-
77.00 - 79.99	C+
73.00 - 76.99	C
70.00 - 72.99	C-
67.00 - 69.99	D+
63.00 - 66.99	D
60.00 - 62.99	D-
<= 59.99	F

Presentations: Each student needs to sign up to present recent research papers and/or hands-on tools relevant to the week's topic; the general model is that the first lecture in the week will be by the instructor providing a survey of the topic area, and the second lecture will have student presentations (2-3 per class) on specific subtopics. The student presentation should convey the underlying technology clearly, situate it in the broader design-automation landscape, and prompt discussion. Sign-ups occur after the class roster is settled.

Projects: Both the mini project and the final project are completed in teams of 2 students. The mini project focuses on the classical model-based design automation workflow (MBSE, MDAO, and basic simulation) applied to a small-scale CPS, documented in a written report. The final project addresses a more ambitious AI-assisted design problem drawn from the second half of the course, documented in a written report and a recorded presentation. The details of the mini and final projects, deliverable expectations, and timeline will be posted on the course site.

Schedule and topics: This schedule is flexible and may be adjusted based on student interests and the emergence of new research. Practical exercises will be integrated into relevant weeks.

Week	Topic
1	Introduction to CPS and Design Automation
2	MBSE Fundamentals Model-Based Systems Engineering (MBSE) is the foundational practice of developing a set of interconnected digital models to represent a system throughout its lifecycle. Instead of relying on static, disconnected documents (like PDFs and spreadsheets), MBSE uses a central, authoritative model as the "single source of

	truth." This topic would introduce core concepts like requirements, behavior, structure, and parameters (V&V), and use a standard modeling language like SysML (Systems Modeling Language) to show how these concepts are formally captured.
3	Architecture Modeling This topic applies MBSE principles to define the high-level structure of a system. Architecture modeling involves identifying the primary components, defining their responsibilities, and mapping the interfaces and interactions between them. Students would learn how to create different architectural "views" (e.g., logical, physical, behavioral) to satisfy various stakeholder concerns and ensure the system's components will integrate correctly.
4	Design Space Modeling The "design space" is the vast, multi-dimensional set of all possible design choices for a system. This includes component selections, software algorithms, and architectural trade-offs. This topic covers how to formally model this space, defining its parameters, constraints (e.g., cost < \$100, weight < 2kg), and metrics for success (e.g., maximize battery life). A well-defined design space is essential for systematic exploration and optimization.
5	Hybrid Dynamics Analysis (Hybrid ODE-based) Cyber-Physical Systems are inherently hybrid systems; their behavior is a mix of continuous physics (like a car's velocity) and discrete events (like a gear shift or a software command). This topic introduces the formalisms and tools and mathematical modeling of the dynamic behavior of such systems. Ordinary Differential Equations (ODEs) are used to describe the continuous dynamics (how variables change smoothly over time), forming the basis for simulation and analysis of the system's physical behavior.
6	Higher Fidelity Analysis (PDE-based) While ODEs are great for system-level dynamics, they often simplify the underlying physics. For detailed analysis of specific components, higher-fidelity models are needed. This topic introduces methods and tools to model complex physical phenomena distributed over space and time, such as fluid dynamics (CFD), heat transfer, or structural stress (FEA). The simulations involved are highly accurate but computationally very expensive.
7	Multi-Disciplinary Analysis & Optimization (MDAO) Real-world systems involve a tight coupling of multiple engineering disciplines (e.g., mechanical, electrical, thermal, software). MDAO is a methodology for analyzing how these different domains interact and for making design trade-offs between them. For example, making an electronic component faster (electrical) might make it run hotter (thermal), requiring a larger heat sink (mechanical). MDAO techniques will be used to find a system-level optimal design that balances these competing objectives
8	Mini Project This would be a hands-on project where students apply the concepts learned so far (MBSE, MDAO, basic simulation) to model and analyze a small-scale Cyber-Physical System. The goal is to solidify understanding of the "classical" design automation workflow before introducing advanced AI techniques.
9	Surrogate Modeling

	High-fidelity PDE simulations (like CFD) can take hours or days to run, making them impractical for large-scale design space exploration. A surrogate model (or metamodel) is an AI-driven approximation of one of these expensive simulations. By training a machine learning model (e.g., a neural network) on data from the high-fidelity simulation, you can create a surrogate that provides near-instantaneous results, enabling rapid design iteration and optimization.
10	Physics-informed Neural Networks (PINNs) PINNs are AI models that embed physical laws (in the form of PDEs) directly into the neural network's training process. Unlike purely data-driven surrogates, PINNs are constrained by the laws of physics. This allows them to make accurate predictions even in regions where training data is sparse and ensures their outputs are physically plausible.
11	AI-assisted Design Space Exploration (DSE) With a vast design space and fast surrogate models, the next challenge is to efficiently search for the best possible designs. This is where AI-assisted DSE comes in. This topic will introduce advanced optimization and search algorithms (e.g., Bayesian optimization, active learning-based, reinforcement learning) that are AI-based or augmented with AI to intelligently navigate the design space to find novel, high-performing solutions much more effectively than manual trial-and-error or brute-force methods
12	LLMs in Design This topic explores the role of Large Language Models (LLMs) like GPT-n in engineering design. The applications include: requirements engineering – translating natural language requirements into formal specifications; code and model generation - simulation and behavior code generation, component modeling and architecture generation; knowledge extraction - parsing technical documents, datasheets, and design knowledge. This is a rapidly emerging area with new applications and startups emerging on a regular cadence, the students will join in exploring and presenting to their colleagues exciting applications and tools
13	Agentic AI frameworks This moves beyond using AI as a singular tool in the context of a design to using agentic framework to define complex and agile design workflow, creating and using AI “agents” as well as “tools”. The topic will introduce state-of-the-art agentic frameworks such as LangGraph, AutoGen, OpenHands etc.
14	Agentic Toolchain for Design This topic will approach practical implementation of the agentic framework concept. An agentic toolchain is the integrated set of software, APIs, simulation tools, and models that the AI agents use to perform their tasks. This module would cover the engineering aspects of building such a system: how to give agents access to tools like CAD software or simulation engines, how to manage their communication, and how to orchestrate the overall workflow.

Late Work: Due dates for presentations and project deliverables will be posted on the course site. Unless noted otherwise, late submissions are accepted for up to three days with a penalty of 10% per day; after three days, work is not accepted without a prior arrangement. If you anticipate difficulty meeting a deadline, please contact the instructor in advance to request an extension.

Students with Disabilities & Access: Vanderbilt is committed to equal access for all students. Students who require accommodations should register with the Student Access Services office (<https://www.vanderbilt.edu/student-access/>) and provide the instructor with their accommodation letter as early in the term as possible, so that arrangements can be made in a timely and confidential manner.

Student Wellbeing: Your wellbeing is important to your success in this course. If you experience stress, or personal or academic challenges that affect your work, you are encouraged to reach out to the instructor and to Vanderbilt's Student Care Network (<https://www.vanderbilt.edu/student-care-network/>), which coordinates the University Counseling Center, the Center for Student Wellbeing, and Student Health. Seeking support is a sign of strength.